

IIT Kharagpur

Department of Mathematics

Advanced Machine Learning

(LTP- 3-0-0,CRD- 3, 40hrs)

Prerequisites: Fundamentals of AI & ML, Design & Analysis of Algorithms

Syllabus :

1. ANN and backpropagation revisited - 3 lectures
2. Deep Learning and Generalization methods - 4 lectures
3. Architecture of Recurrent Neural Networks - 3 lectures
4. Training RNNs - 3 lectures
5. Echo-State Networks, Long Short-Term Memories, and Gated Recurrent Units - 4 lectures
6. Architecture of Convolutional Neural Networks - 3 lectures
7. Training CNNs - 3 lectures
8. Case studies and applications of CNNs - 3 lectures
9. Graph Neural Networks - 3 lectures
10. Gated GNNs, comparison with image CNNs, backpropagation in GNNs and applications - 3 lectures
11. Reinforcement Learning: Bandit algorithms, Temporal Difference methods, Dynamic Programming methods - 3 lectures
12. Deep Reinforcement Learning - 3 lectures
13. Monte Carlo Sample, policy gradient methods, Monte Carlo Tree Search - 3 lectures

Total: 35 lectures

References:

1. Pattern recognition and machine learning, Christopher M. Bishop, Springer, 2007.
2. Neural Networks and Deep Learning - A Textbook, Charu C. Aggarwal, Springer, 2018.
3. Deep Reinforcement Learning, Aske Plat, Springer, 2022.
4. Recurrent Neural Networks - From Simple to Gated Architectures, Fathi M. Salem, Springer, 2022.
5. Graph Neural Networks: Foundations, Frontiers, and Applications, Lingfei Wu, Peng Cui, Jian Pei, Liang Zhao (Eds.), Springer 2022.

Instructors: B Roy, B Banerjee, S Khare, A Banerjee